

Henry Asante

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Education

Grambling State University

Graduating May 2028

BSc. Computer Science, GPA: 4.0/4.0

Grambling, Louisiana

Coursework: Deep Learning, Operating Systems, Artificial Intelligence, Data Structures, Object-Oriented Design

Skills

- **Languages:** Python, JavaScript, C/C++, Java
- **Infra & Dev Tools:** Git, CI/CD, Linux, Jenkins, GitHub Actions, MongoDB, Docker, AWS,
- **AI/ML & Data:** SQL, PyTorch, TensorFlow, Scikit-learn, RAG

Experience

Allstate Insurance Company

Northbrook, Illinois

Software Engineer Intern

May 2026 – August 2026

- Automated regression testing for the auto insurance user onboarding workflows using Playwright and JavaScript.
- Improved automation reliability by writing reusable end-to-end tests and resolving flaky pipeline failures.
- Accelerated continuous testing by integrating GitHub Actions, and Jenkins into CI/CD workflows.

Dell Technologies

Round Rock, Texas

Software Engineer Intern

May 2025 – August 2025

- Built an AI testcase recommendation platform with LLMs, embeddings, & semantic similarity for regression analysis.
- Reduced testcase selection time significantly by mapping 5,000+ test cases in qTest to 1,200+ Jira defects.
- Delivered an interactive engineering dashboard using React & Flask to visualize AI-powered regression recommendations
- Improved recommendation quality by 10% via embedding optimization reaching 96% accuracy.

AmaliTech NSP Training

Remote

Software Developer Trainee

May 2024 – August 2024

- Delivered a scalable learning platform using Spring Boot REST APIs serving over 15,000 educational Q&A pairs.
- Reduced backend latency by approximately 40% through SQL optimization, Redis caching, and multithreading.
- Improved software reliability by implementing JUnit and Mockito tests with over 95% code coverage.

KNUST Engineering Lab

Kumasi, Ghana

Full Stack Engineering Intern

August 2023 – November 2023

- Developed a cross-platform engineering simulation platform using Vue.js, Python, TypeScript, and Electron.
- Enabled engineering simulations via RESTful backend services connecting user interfaces with computation pipelines.
- Enhanced engineering support by integrating a RAG-powered assistant for circuit troubleshooting and design guidance.

Projects

IntelliJobs

- Built a full-stack web & mobile jobs platform using React, Flutter, and containerized FastAPI services with PostgreSQL.
- Designed core backend workflows for resume ingestion, personalized recommendations, & apps tracking with JWT auth.
- Implemented a multi-source discovery pipeline aggregating listings from 4 ATS platforms with resume-aware ranking.
- Engineered a multi-service architecture using API-first practices, schema migrations, and 1,500+ automated tests.

Orbit-sh: A Modular Unix Shell with Systems Integration

[view project](#)

- Built a Unix shell in C supporting command parsing, foreground/background execution, & built-in commands.
- Implemented pipeline support (`|`) using `pipe()` and `dup2()` for inter-process communication between child processes.
- Prevented zombie processes using `sigaction(SIGCHLD)` w/ non-blocking `waitpid()`, ensuring child process reaping.

Goal Misgeneralization in a Model-Based Agent

[view project](#)

- Engineered a physics-based simulation pipeline in a 4D continuous state space to evaluate autonomous agent robustness
- Integrated a neural dynamics model with a CEM planner for 200-step trajectory rollout and receding-horizon MPC.
- Built a reproducible evaluation framework across 4 environment shifts with up to $2\times$ changes in mass and friction.